

OLAP
preparation
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1) City Hospital Data Warehouse

a) Snowflake Schema

Fact Table:- Visit-Fact

Measures:-

- Visit-Count
- Avg-Charge

Dimension Tables:-

- > Patient: Patient-ID, Name, Age, Gender
- > Doctor: Doctor-ID, Name, Specialization
- > Department: Department-ID, Department-Name, Hospital-Block
- > Time: Day -> Month -> Quarter -> Year

The Fact table is connected to all four dimension tables, forming a snowflake schema with normalized dimensions.

b) OLAP Operations

To list the average consultation fee department-wise for each year:

1. Start with the base ~~consultation fee~~ cuboid {Patient, Doctor, Department, Time}

2. Roll up the time dimension from Day to Year

3. Roll-up Patient to All.
4. Roll-up Doctor to All.
5. Keep the Department dimension unchanged.
6. Calculate the average value of Avg-charge.

The Final result is Department-wise Average consultation Fee For each Year.

c) number of cuboids

number of dimensions = 4

Levels in each dimension = 4

Formula:-

$$\text{Total cuboids} = 4 \times 4 \times 4 \times 4 = 4^4 = 256$$

Answer 256 cuboids.

25)

Smart city Project

Star Schema Design

Fact Table: smartcity - fact

Measures:-

- > vehicle count
- > congestion level.
- > energy consumption.
- > Air quality Index
- > noise level
- > Ridership
- > Punctuality Percentage.
- > citizen satisfaction score.

Dimension Table:-

Traffic Dimension

- > sensor ID
- > Road name
- > Area.

Time Dimension

- > Day
- > Month
- > Quarter
- > Year.

Transport Dimension

- > Route number
- > vehicle type

energy Dimension

- > Device type
- > Zone

citizen Dimension

- > Feedback type
- > Location
- > Age Group

Justification

The Fact table stores numerical values, while dimension table describe the data. A star schema improves query performance and supports efficient OLAP analysis.

8) Banking Data Warehouse

a) Galaxy schema

The banking warehouse contains two fact tables:

Transaction-Fact

→ Transaction-Count

→ Amount.

Loan-Fact

→ Loan-Amount

→ Interest.

Both fact tables share the following dimensions:

→ Customer

→ Branch

→ Time

→ Account-type.

This structure is called a Galaxy schema (Fact constellation schema).

1. Roll-up the time dimension from day to Year
2. keep the Branch dimension.
3. calculate:
 - Average Transaction Amount.
 - Average Loan Amount.

The final report displays Branch-wise Average Transaction Amount and Loan Amount for each Year.

c) number of cuboids.

Dimensions = 4

Levels = 4

Total cuboids = $4^4 = 256$

• Answer: 256 cuboids.

4) Manufacturing company

1) Roll-up

Roll-up product dimension from Batch $\rightarrow$ type $\rightarrow$ category and the time dimension from week $\rightarrow$ Month $\rightarrow$ Quarter $\rightarrow$ Year.

This summarizes Defect Rates by product category for the current Year across all Factories.

2) Drill-down

Select one Factory and drill down from Factory $\rightarrow$ Department $\rightarrow$ Product type.

This provides a detailed analysis of Defect Rates.

3) Slice

Apply the condition Supplier Rating ≥ 4 .

Display product units across all regions.

4) Dice

Select :-

→ Product categories

→ Factory Locations

Analyses :-

→ Defect Rate

→ Supplier Reliability Percentage

This provides a detailed multidimensional analysis.

5) Retail Chain

1) Roll-up

→ Product: SKU → Subcategory → Category

→ Store: store ID → City → Region.

→ Time: Day → Month → Quarter.

The summarizes inventory turnover by Product category across all stores for the last quarter.

2) Drill-down

Select the electronics category in a specific region.

Drill down from:

category → subcategory → sku.

This gives a detailed analysis of Sales.

3) Slice

Apply the conditions.

→ Gender = Male

→ Membership Tier = Bronze

Display Total Sales Revenue across all Stores.

4) Dice

Select

→ specific cities

→ Product categories

→ Time Periods.

Analyse:-

→ Total Sales Revenue

→ units sold.

